

BUIzz

BUIzz - Mingi Jung, Donggyu Kim, Mijung Kim, Seongil Wi, GitHub.

- Published: date not stated
- Original: <<https://github.com/WebSec-Lab/BUIzz>>
- Preserved from: <https://github.com/WebSec-Lab/BUIzz> (git) on 2026-08-19
- Repository commit: 4d573c437f0ca88824b9a15af66255ad147a5f49
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Content

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This reference is a source-code repository. The archive preserves its documentation at an exact commit; the code itself stays in a private mirror and is never checked out, built or run.

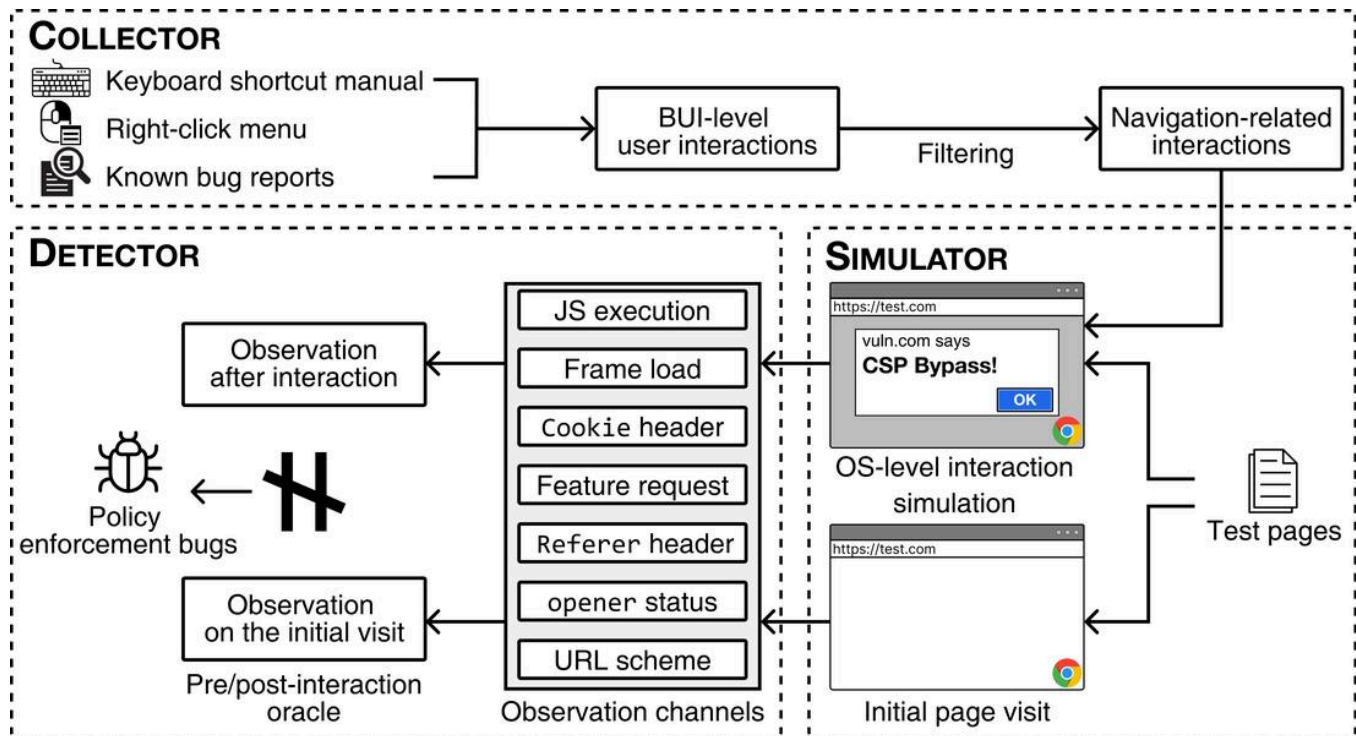
- Repository: <<https://github.com/WebSec-Lab/BUIzz>>
- Commit: 4d573c437f0ca88824b9a15af66255ad147a5f49
- Documents preserved: 2

README.md

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BUIzz

BUIzz is a browser testing framework designed to find policy enforcement bugs by simulating BUI (Browser User Interface)-level user interactions at the OS level.



Repository Structure

BUIZZ/

```
crawler/      # User interaction study data
example/      # Minimal working example
fuzzer/       # Core fuzzer
  fuzzer.py    # Entry point
  user_scenario_gen.py # Scenario generator
lib/          # Browser drivers & interaction helpers
scenario/     # Generated scenario JSON files
test_list/    # Corpus URL lists per policy
browser_info/ # Browser executable paths & UI configuration
server/       # Per-policy report servers
  samesite/
  csp/
  coop/
  hsts/
  pp/
  rp/
  sandbox/
  xfo/
fuzzerV2/     # Extended fuzzer variant
scheduler/    # Distributed fuzzing coordinator
bugs/         # Analyzer output
Figure/       # Paper figures
safe_error/   # Baseline-error URL list
analyzer.py   # Inconsistency analyzer
base_line.py  # Baseline collector
base_lineV2.py # Extended baseline
deduping.py   # Bug deduplicator
makeDB.py     # Database initializer
schema.sql    # Database schema
setup.ps1     # Environment setup
certs.ps1     # TLS certificate generation
```

Tested Environments

The artifact was developed and validated on the following setup:

Component	Tested Version
OS	Windows 11 23H2 (64-bit)
Python	3.11.x
Playwright	1.44+
MySQL	8.0.x

Component	Tested Version
Docker Desktop	4.x
Chrome	138
Firefox	139
Edge	136
Opera	118
Brave	1.79
Whale	4.32

Hardware: 16 GB RAM minimum, 50 GB free disk space recommended

Prerequisites

The following programs must be installed before running BUIZZ:

Program	Version	Purpose
Windows 10/11	64-bit	Required OS — pywinauto and pyautogui drive real OS-level input
Python	3.11+	All fuzzer, analyzer, and server scripts
Docker Desktop	Latest	Runs the per-policy report servers via <code>docker compose</code>
MySQL	8.0+	Stores test results (<code>event_entry</code> table); must be running on <code>localhost:3306</code> with user <code>root</code> / password <code>1234</code>
mkcert	Latest	Generates locally-trusted TLS certificates (<code>certs.ps1</code> installs it automatically)
Browsers	See below	The actual browsers under test

Browsers required (install only those you intend to test):

Browser	Installer
Google Chrome	https://www.google.com/chrome
Microsoft Edge	Pre-installed on Windows 10/11
Mozilla Firefox	https://www.mozilla.org/firefox
Brave	Portable binary bundled in <code>example/brave-v1.80.120-win32-x64/</code> — no installation needed
Opera	https://www.opera.com
Naver Whale	https://whale.naver.com

Setup

1. Install dependencies

Run `setup.ps1` **as Administrator** in PowerShell. It installs all required Python packages and automatically updates the hosts file with the required domains.

```
# Auto-detect the local IP
PowerShell -ExecutionPolicy Bypass -File setup.ps1

# Specify an IP explicitly (e.g. when running the server on a separate machine)
PowerShell -ExecutionPolicy Bypass -File setup.ps1 10.20.23.182
```

If no IP is provided, the script detects the primary non-loopback IPv4 address of the current machine automatically.

The script installs the following packages and runs `playwright install` :

```
playwright psutil pywinauto pyautogui webdriver-manager mysql-connector-python
```

Manual alternative — if you prefer to install packages individually: `` bash pip install playwright psutil pywinauto pyautogui webdriver-manager mysql-connector-python python -m playwright install ` Then add the following entries to `C:\Windows\System32\drivers\etc\hosts` : ` <your-ip> leak.test <your-ip> addition.com <your-ip> attacker.test <your-ip> attacker.com <your-ip> victim.com ``

2. Generate TLS certificates

Run `certs.ps1` to install mkcert and generate TLS certificates for all HTTPS servers:

```
PowerShell -ExecutionPolicy Bypass -File certs.ps1
```

3. Create the database

```
python makeDB.py
```

Collector

The `crawler` directory contains an XLSX file that comprehensively collects the user interactions gathered in our study.

Simulator

Server

Navigate to the target policy's server directory and start it using Docker Compose:

```
cd server/samesite
docker compose up
```

Available server directories: `csp`, `samesite`, `pp`, `coop`, `hsts`, `rp`, `xfo`, `sandbox`

Scenario Generation

User interaction scenarios are generated using `user_scenario_gen.py` in the `fuzzer` directory.

```
python fuzzer/user_scenario_gen.py -s samesite -b chrome -d 1
```

Option	Description
<code>-b</code>	Browser (<code>chrome</code> , <code>firefox</code> , <code>edge</code> , <code>opera</code> , <code>brave</code> , <code>whale</code>)
<code>-s</code>	Security policy (<code>samesite</code> , <code>csp</code> , <code>sandbox</code> , <code>pp</code> , <code>coop</code> , <code>hsts</code> , <code>rp</code> , <code>xfo</code>)
<code>-d</code>	Depth (<code>1</code> = single interaction, <code>2</code> = two-interaction combination)

OS-Level Simulation

The fuzzer is executed via `fuzzer.py` :

```
python fuzzer/fuzzer.py -s samesite -b chrome
```

Detector

Baseline collection

To record the pre-interaction state, run `base_line.py` to capture baseline enforcement behavior before any user interaction is performed:

```
python base_line.py -s samesite -b chrome
```

Analysis

Run `analyzer.py` to compare pre-interaction and post-interaction enforcement outcomes and identify inconsistencies:

```
python analyzer.py -s samesite -b chrome
```

Add `--lenient` for higher recall (may increase noise):

```
python analyzer.py -s samesite -b chrome --lenient
```

Results are written to `bugs/<policy>/interaction_diff_<browser>.txt`.

Deduplication

Run `deduping.py` to deduplicate flagged inconsistencies and group them into distinct root-cause bugs:

```
python deduping.py -s samesite
python deduping.py -s samesite -b chrome    # single browser
python deduping.py -s samesite --merge-tags  # merge by root cause
python deduping.py -s samesite -d 1         # depth-1 scenarios only
python deduping.py -s samesite -d 2         # depth-2 scenarios only
```

Database Schema

The `event_entry` table stores all fuzzing and baseline records. See `schema.sql` for the full annotated schema, or run `python makeDB.py` to create the database automatically.

Column	Description
<code>browser_name</code>	Browser under test (<code>chrome</code> , <code>firefox</code> , <code>brave</code> , ...)
<code>scenario_id</code>	Scenario file name (e.g. <code>0_DEPTH1.json</code>); NULL for baseline records
<code>corpus</code>	Full URL of the corpus page loaded
<code>event_type</code>	<code>corpus</code> (baseline) or <code>interaction</code> (post-interaction result)
<code>corpus_type</code>	Security policy (<code>samesite</code> , <code>csp</code> , <code>referrer-policy</code> , ...)
<code>leak</code>	Leak channel that triggered the report (e.g. <code>a-href</code> , <code>fetch</code> , <code>img</code>)
<code>violation</code>	Enforcement outcome reported by the server (e.g. <code>lax</code> , <code>1</code> , <code>empty</code>)
<code>interaction</code>	Human-readable label of the simulated interaction; NULL for baseline
<code>timestamp</code>	UTC timestamp of record insertion

Citation

You can cite our paper with the following bibtex entry.

```
@INPROCEEDINGS{jung:usenixsec:2026,  
  author = {Jung, Mingi and Kim, Donggyu and Kim, Mijung and Wi, Seongil},  
  title = {{BUIzz}: Finding Policy Enforcement Bugs via Interaction Simulation on the Browser User Interface},  
  booktitle = {In Proceedings of the {USENIX} Security Symposium},  
  pages = {4961--4980},  
  year = 2026  
}
```

example/README.md

Blob 4151e40e1377, 7330 bytes, at commit 4d573c437f0c.

Minimal Working Example — Brave

This directory provides a self-contained environment to run a complete BUIZZ pipeline using **Brave** without installing any additional browser.

[!IMPORTANT] Every command in this document must be run from the **example/** directory. Open a terminal, `cd` into `example/`, and keep it there for all steps. `` powershell cd path\to\BUIZZ\example ``

Directory Structure

```
example/  
  setup.py          # Downloads Brave + seeds the pre-configured profile  
  brave-profile-template/ # Pre-configured Brave profile (Shields off, English UI)  
  mini_scenario.py  # Deploys pre-selected bug scenarios into fuzzer/scenario/  
  samesite_split/   # Pre-selected scenarios for SameSite split-view bug (bug_20)  
  csp_split_blob/   # Pre-selected scenarios for CSP split-view blob: bug (bug_07)  
  csp_split_data/   # Pre-selected scenarios for CSP split-view data: bug (bug_06)
```

Why Brave?

Among the six browsers tested, Brave is the only one suitable for a pinned reproducible example. Chrome, Firefox, and Opera auto-update on launch; Edge and Whale do not offer older version archives on their official sites. Brave alone provides versioned standalone ZIP archives on GitHub Releases with no auto-update.

Prerequisites

1. Install Python packages and configure hosts file (run as Administrator)

```
PowerShell -ExecutionPolicy Bypass -File ..\setup.ps1
```

2. Generate TLS certificates

```
PowerShell -ExecutionPolicy Bypass -File ..\certs.ps1
```

3. Create the MySQL database

```
python ..\makeDB.py
```

4. Download and configure the portable Brave binary

```
python setup.py
```

`setup.py` does three things:

1. Downloads `brave-v1.80.120-win32-x64.zip` (~200 MB) from the official Brave GitHub release and extracts it.

1. Seeds a **pre-configured, isolated Brave profile** at `example/brave-profile/`

by copying `brave-profile-template/` — Shields already disabled and the UI forced to English. (Skipped if `brave-profile/` already exists, so re-running never wipes your state.)

1. Writes the launch command into `..\fuzzer\browser_info\browser_info.json` , pinning the extracted `brave.exe` , the dedicated `--user-data-dir` , and `--lang=en-US` .

Because Brave runs from its own `--user-data-dir` , it never collides with any system-installed Brave, and **no manual browser configuration is required**.

Docker Desktop must be running before Step 1 below.

Brave configuration (pre-applied — no action needed)

Brave's built-in Shields would otherwise interfere with the fuzzer's cross-site requests, so they must be turned off. **You don't need to do this manually** — the bundled profile in `brave-profile-template/` already has the required settings, and `setup.py` copies it into `example/brave-profile/` . The fuzzer then launches Brave against that profile.

The pre-applied settings are:

Setting	Location	Value
Trackers & ads blocking	<code>brave://settings/shields</code>	Disabled
Upgrade connections to HTTPS	<code>brave://settings/shields</code>	Disabled (prevents forced HTTPS redirect on local test domains)
Block fingerprinting	<code>brave://settings/shields</code>	Disabled
Block cookies	<code>brave://settings/shields</code>	Allow all cookies

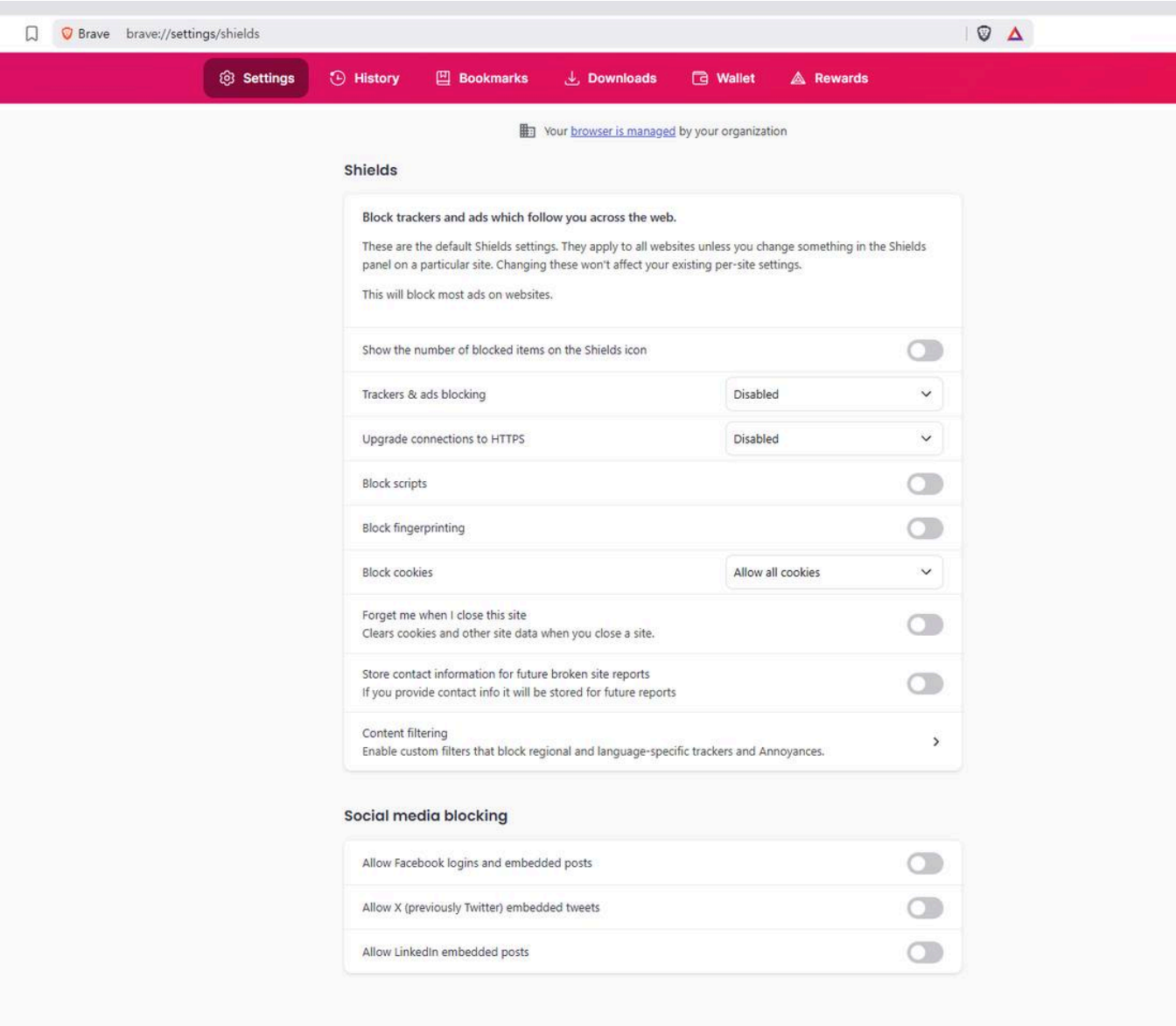
Setting	Location	Value
Startup behavior	brave://settings/onStartup	Open the New Tab page
Display language	brave://settings/languages	English (United States)

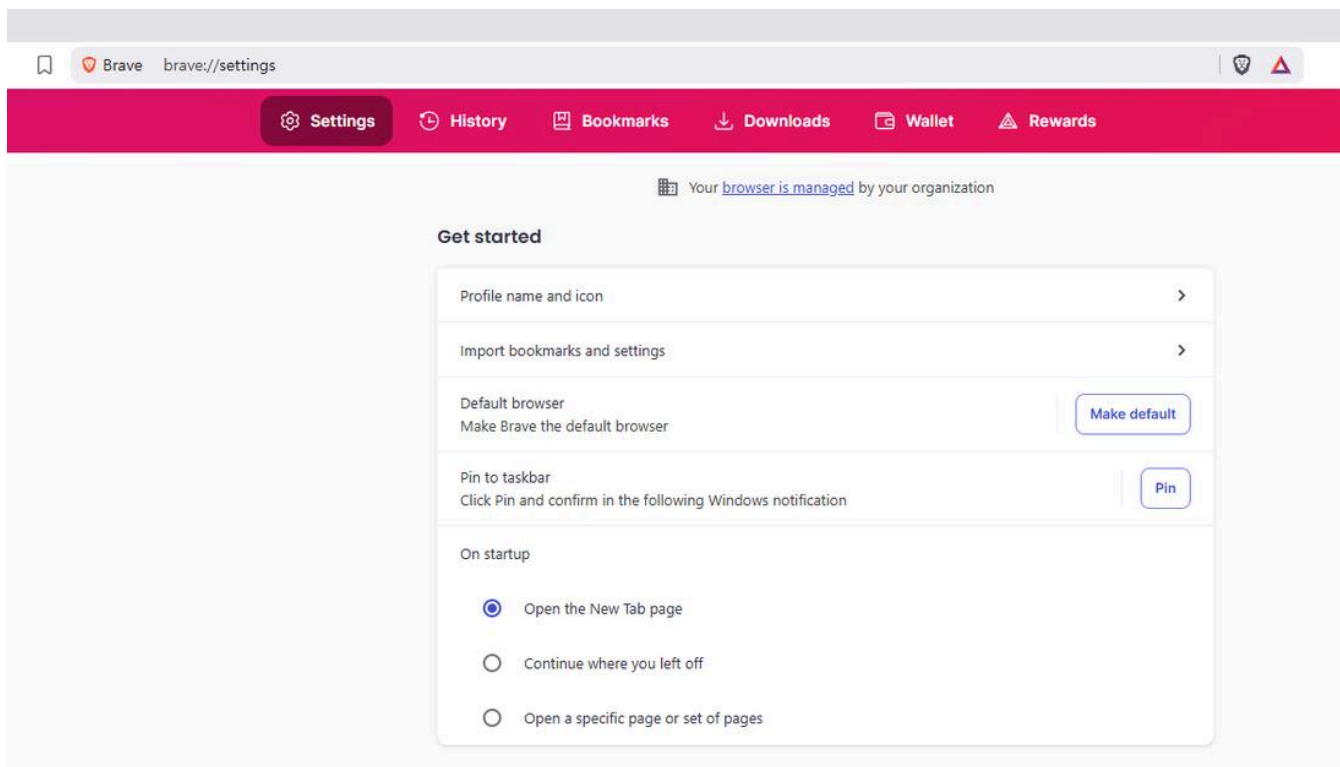
<details> <summary>Optional — set these manually instead</summary>

If you prefer to configure Brave yourself, launch the bundled binary with its dedicated profile:

```
.\brave-v1.80.120-win32-x64\brave.exe --user-data-dir=".\\brave-profile" --lang=en-US --no-first-run
```

Then apply the values in the table above under brave://settings/shields and brave://settings/onStartup :





</details>

Step 1 — Deploy pre-selected scenarios

Choose which bug to test and run the corresponding command:

Argument	Policy	Bug
1	SameSite	Split-view bug (bug_20)
2	CSP	Split-view blob: bug (bug_07)
3	CSP	Split-view data: bug (bug_06)

```
python mini_scenario.py 1 # SameSite split-view bug (bug_20)
python mini_scenario.py 2 # CSP split-view blob: bug (bug_07)
python mini_scenario.py 3 # CSP split-view data: bug (bug_06)
```

The remaining steps use the policy that matches your choice: `samesite` for argument 1, `csp1` for arguments 2 or 3.

Step 2 — Start the report server

```
# For SameSite (scenario 1)
cd ..\server\samesite\corpus && docker compose up -d && cd ..\..\example

# For CSP (scenario 2 or 3)
cd ..\server\csp && docker compose up -d && cd ..\..\example
```

Step 3 — Collect the baseline

```
# SameSite
python scenario_base_line.py -b brave -s samesite

# CSP
python scenario_base_line.py -b brave -s csp1
```

`scenario_base_line.py` reads the scenario files already deployed in `fuzzer/scenario/brave/<policy>/` and visits only those corpus URLs using Chrome as the baseline browser (automatically selected).

Step 4 — Run the fuzzer

```
# SameSite
python ..\fuzzer\fuzzer.py -s samesite -b brave

# CSP
python ..\fuzzer\fuzzer.py -s csp1 -b brave
```

Do **not** move the mouse or switch windows while the fuzzer is running.

Step 5 — Analyze inconsistencies

```
# SameSite
python ..\analyzer.py -s samesite -b brave

# CSP
python ..\analyzer.py -s csp -b brave
```

Expected output:

[strict] flagged N/M interaction rows
[+] Results written to bugs/<policy>/interaction_diff_brave.txt

Step 6 — Deduplicate results

```
python ../deduping.py -s samesite -b brave # or -s csp1
```

Expected output:

```
=====
BUIZZ deduplication - policy=samesite browser=brave
=====
Raw inconsistency records : 12
Distinct bugs (after dedup): 4

Bug #01: (Open link in a split window, <a>, https:)
  browsers : brave
  scenario : 8_DEPTH1.json
  records : 3

Bug #02: (Open link in background tab, <a>, https:)
  browsers : brave
  scenario : 1_DEPTH1.json
  records : 5

Bug #03: (Open link in new tab, <a>, https:)
  browsers : brave
  scenario : 11_DEPTH1.json
  records : 2

Bug #04: (Open link in new window, <a>, https:)
  browsers : brave
  scenario : 13_DEPTH1.json
  records : 2

Total distinct bugs: 4
```

The following is a sample record from the analyzer's inconsistency output — an interaction where Brave's behaviour differed from the baseline:

```
{'browser_name': 'brave', 'scenario_id': '8_DEPTH1.json', 'interaction': 'Open link in a split window', ... , 'leak': 'a-href', 'vic
```

The `violation: strict` field means the `SameSite=Strict` cookie was transmitted — a behaviour that should be blocked, confirming a SameSite policy bypass.

This bug has been assigned **CVE-2025-48980**.

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